



Electrical, structural, morphological and photovoltaic properties of Au/n-Ge heterojunctions using V₂O₅ interfacial layer

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ABSTRACT

Influence of vanadium pentoxide (V₂O₅) interfacial layer on electrical, structural, morphological and photovoltaic properties of the Au/V₂O₅/n-Ge Schottky diodes is presented. The analysis of morphological properties of the V₂O₅ thin films on the n-Ge surface using AFM and FESEM with EDS reveals the films are homogeneous. The AFM measurements of V₂O₅ thin films show a fairly smooth surface with RMS roughness of 2.44 nm. Further, the FESEM with EDS measurements of the V₂O₅/n-Ge layers evidenced with V, O and Ge elements signifies the presence of V₂O₅ thin films on the n-Ge substrate. The structural properties of V₂O₅ deposited on the n-Ge substrate were studied from XRD and Raman measurements. These properties of V₂O₅ films reveal that the films were highly crystalline with orthorhombic structure and significant diffraction intensity corresponds to V₂O₅ with various crystal orientations. The electrical properties of the n-Ge heterostructure were studied with and without V₂O₅ thin interlayer. The Schottky barrier parameters such as barrier height and shunt resistance seem to be superior for the V₂O₅/n-Ge heterojunction compared to without V₂O₅ interlayer. The current conduction mechanism of the V₂O₅/n-Ge heterojunction signifies SCLC current transport process assisted with traps is found to be dominant at the V₂O₅-Ge interface preferably at large forward bias voltages. Frequency-dependent capacitance and series resistance of the V₂O₅/n-Ge heterojunction and Au/Ge junction are presented. The photovoltaic property of heterojunction was studied under illumination at 100 mW cm⁻² and the studies show a responsivity of 534 mA W⁻¹ at -1 V. In view of these observations, the V₂O₅ interlayer has demonstrated its effectiveness in tuning the barrier characteristics of metal-Ge Schottky junctions. Its photoresponse to illumination also makes its suitability in solar cell of photovoltaic applications.

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1 Introduction

Germanium (Ge)-based MOS devices have gained larger attention over the past decade due to their technological importance over Si-based technology [1–3]. Si-based MOS devices inherently suffer from low drain currents upon downscaling and also limitation of carrier velocities sets an edge to the scaling of Si-based MOSFET devices. On the other hand, Ge semiconductor offers larger carrier mobility and low effective mass of the carriers which further improves larger carrier velocities and hence smaller gate delay [4]. This sets the motivation for the fabrication and characterization of Ge-based MOS devices using various thin and thick oxide layers over Ge semiconductors and studied its electrical properties. A very serious problem the researchers faced during the early days of fabrication of Ge based two terminal devices (Schottky junctions) is Fermi level pinning near the valence band edge and formation of native oxide (GeO_2) on the Ge surface. A GeO_2 thin layer generally expresses its solubility in water and is unstable to temperature which degrades the device's performance. An initiative has been taken to fabricate Ge MOS devices to remove the Fermi level pinning as well it should also create an effective oxide layer as an interlayer (IL) over Ge substrate. Initially, very thin oxide layers like Al_2O_3 , MgO , ZnO , TiO_2 and Y_2O_3 have been coated on Ge substrate not only to remove Fermi level pinning at the Ge interface but also to engineer the barrier properties [5–13]. But, the focus on the design and electrical characterization of Ge-based MOS devices using thick oxide IL is found to be very limited [14–16].

Highly standardized, well-known crystalline silicon (c-Si) solar cells are under competition from low-cost photovoltaic cell fabrication. The research has accelerated in the last decade to look into the alternatives of crystalline-Si solar cells with bulk heterojunction solar cells. The bulk heterojunction solar cells include the fabrication of a thin interlayer either electron or hole-selective layers on the top of crystalline-Si. The thin interlayers might be either organic or inorganic substances. Various reports are available which utilize electron and hole-selective transition metal oxides (TMOs) as an interlayer in the fabrication of c-Si bulk heterojunction solar cells [17–19]. Among the various TMOs, hole selective materials such as MoO_3 , V_2O_5 and WO_3 are featured with large work function (> 5 eV), reduced interface state density, optimum transparency and also observed to be a passivating

layer to c-Si crystalline solar cells. Among the hole selective layers mentioned V_2O_5 proved to be the most passivating layer to c-Si heterojunction solar cells than other TMOs [17, 20]. Vanadium oxides made the subject of number of studies, mainly due to their interesting physical properties near ambient pressure and temperature conditions, which makes V–O films attractive for applications in smart window, thermochromic applications, energy storage and optoelectronic switching devices [21–23].

V_2O_5 modified c-Si heterojunction solar cells have been widely studied over the past decade due to their economical viability and custom design [17, 24–30]. The physical and electrical properties of nitrogen ion diffused V_2O_5 thin films on different substrates were studied [30]. Upon increasing the nitrogen influence a decrease in optical bandgap and electrical conduction is observed in V_2O_5 thin films due to the creation of oxygen vacancies and traps. A comparative study has been carried out in the fabrication of p–n heterojunction solar cells with TMOs (MoO_3 , WO_3 and V_2O_5) as p-type front contacts on the top of n-type c-Si [17]. The results reveal larger conversion efficiency for $\text{V}_2\text{O}_5/\text{Si}$ heterojunctions than other MoO_3 , WO_3 -based heterojunctions. Garcia-Hernansanz et al. [27] realized a MoO_3 -based crystalline Si solar cell and studied the different current transport processes. Bias-dependent current transport processes were observed in this heterojunction, at low bias the hole carriers tunnel through the gaps states provided by the MoO_3 interlayer and at larger bias a high barrier height was created due to the larger work function of the MoO_3 interlayer. $\text{V}_2\text{O}_5/4\text{H-SiC}$ heterojunction-based solar-blind photodetectors were fabricated which found their suitability for cryogenic applications [25]. V_2O_5 nanorods were grown on n-Si in a metal–semiconductor–metal (MSM) structure and studied its photodiode properties [26]. They observed high performance of photodetection properties under illumination due to the excellent crystalline quality of the V_2O_5 thin interlayer. Temperature-dependent electrical properties of $\text{V}_2\text{O}_5/4\text{H-SiC}$ junctions were analysed by Bellone et al. [24]. The TMO interlayer causes to transport of the carriers across the junction and is dominated by field-influenced thermion emission (TE). Kaya et al. [29] studied the physical and electrical properties of $\text{V}_2\text{O}_5/\text{n-Si}$ heterojunctions with varied thicknesses of V_2O_5 from 100 to 400 nm. The variation of resistivity and optical properties of the V_2O_5 thin films were discussed at different thicknesses. The electrical

properties observed for V_2O_5 /n-Si reveal its applications in optoelectronic devices. Rare-earth doped V_2O_5 nanorods modified n-Si heterojunctions were fabricated and characterized the optical, structural, morphological and electrical properties of heterojunctions [28]. These MIS structures show enhanced photoresponsivity under illumination compared to dark. This shows that the V_2O_5 substances can be utilized as photo absorbers in various photovoltaic applications.

Very few recent reports are available on the fabrication and electrical, photovoltaic characterization of Ge heterojunctions using a hole-selective TMO layer [15, 31–33]. The report submitted by Alcaniz et al. [15] describes the formation of MoO_x interfaces with Ge substrate and opens the doors for the fabrication of efficient photovoltaic devices with new device structures. Electrical properties of GeO_x passivated Ge MSM photodiode was investigated by Kang et al. [31]. The GeO_x interlayer in Ni/n-Ge Schottky diode shows significant reduction in the reverse current by one order of magnitude than compared with Al_2O_3 interlayer due to decreased surface leakage current and trap density at the interlayer-Ge interface. Lee et al. [32] modified Ni/n-Ge Schottky structures with Ta-oxide interlayer and studied its electrical properties. The significant improvement in electrical properties (increased barrier height and lower reverse leakage current) of n-Ge Schottky structures with Ta-oxide interlayer could possibly attributed to effective passivation of Ge surface with Ta-oxide interlayer and significant reduction in effective surface state density at the oxide-Ge interface. Recently, ZnO interlayer modified Pt/Ge MOS junctions were demonstrated by Kim et al. [33]. The ZnO interlayer shown improvement in the forward current characteristics which enhances the barrier height of Pt/ZnO/Ge junction than compared to without interlayer.

Considering the advantages of TMOs-based c-Si bulk heterojunction solar cells and the suitability of Ge over Si because of high carrier mobility will provide an effective aptness for scaling and broad spectral range for the design of highly efficient photodiode or photovoltaic structures. This makes us to fabricate V_2O_5 /Ge heterojunctions and to study their physical and optoelectronic properties. In the present work, V_2O_5 thin film was fabricated on quartz glass at a deposition temperature [substrate temperature (T_s)] of 250 °C to study its physical properties such as optical, and Raman measurements. Similarly, with the same experimental conditions, V_2O_5 thin film was coated on an

n-Ge substrate to study its structural (XRD measurements), morphological (AFM and FESEM), electrical (I - V and C - V) and photo-electrical measurements (I - V under illumination). The highly rectifying nature of V_2O_5 /Ge heterojunctions was realized compared to without V_2O_5 IL. The various barrier properties such as barrier height (Φ_b), ideality factor (n), series resistance (R_s) and shunt resistance (R_{sh}) were evaluated using forward bias current-voltage (I - V) properties of the contacts. Enhanced barrier properties of V_2O_5 /Ge heterojunctions reveal significant passivation of the V_2O_5 layer and its noteworthy band engineering between IL and Ge substrate. Further, these junctions were characterized using frequency-dependent capacitance-voltage (C - V) measurements. The frequency-dependent capacitance and series resistance were studied for varied bias potential. Lastly, the photovoltaic properties of the V_2O_5 /Ge heterojunctions were analysed under dark and illumination at 1 sun condition with 100 mW cm⁻² optical power. Various photosensitive properties of heterojunctions such as sensitivity or photo-to-dark current ratio (PDCR) and responsivity were evaluated.

2 Experimental details

In the present investigation, n-Ge (Sb doped) substrates with a doping concentration of 5×10^{15} cm⁻³ were utilized for the fabrication of Ge-based heterojunction with V_2O_5 thin interlayer deposited at substrate temperature (T_s) of 250 °C. Initially, the substrate was degreased in organic solvents for 5 min. each followed by drying in N₂ ambient. After cleaning, the samples were loaded in a vacuum system to deposit ohmic contact (Al) on the rough side of the sample. The thickness of Al was found to be 50 nm as determined using a thickness monitor. After this, the sample surface was changed to a polished side to deposit the interlayer (V_2O_5) using thermal evaporation. The V_2O_5 powder was purchased from Sigma-Aldrich with CAS no. 1314-62-1, having a purity of > 99.6%. By keeping the deposition/substrate temperature (T_s) to 250 °C, the V_2O_5 powder was evaporated thermally onto the smooth surface of the Ge substrate. V_2O_5 -coated Ge substrates were removed and placed on a circular shadow mask with a 0.5 mm diameter to deposit Schottky contact (Au). The thickness of Schottky metal was found to be 40 nm. A similar procedure was adopted to fabricate Au/Ge contact

without interlayer for comparison. All the vacuum depositions were carried out in vacuum coating unit model 12A4-D (Hind Hivac system) at the base pressure of 4×10^{-6} mbar. V_2O_5 thin layers were also coated on glass substrates to perform optical (JASCO, V-670 model) and Raman measurements (Horiba, Model: LabRam HR Evolution). Similarly, V_2O_5 /Ge layers were prepared to study the morphological properties like AFM (Nanosurf, Switzerland, Model: Nanosurf AFM) and FESEM with EDS, cross-sectional FESEM (Thermo Fisher, Model: FEI QUANTA 250 FEG). Finally, the electrical properties Au/ V_2O_5 /Ge/Al heterojunction and Au/Ge/Al Schottky junction was studied using current–voltage (I – V) properties with Keithley source meter (Model No. 2400) and capacitance–frequency (C – f) measurements were performed with an LCR meter (model No PSM 1700). The photovoltaic properties of V_2O_5 /Ge heterojunction was also studied using a solar simulator (PECCELL Inc. Japan, PEC-L01) and source meter (Keithley, 2401, USA).

3 Results and discussion

3.1 Optical, structural and morphological properties

The optical properties of V_2O_5 thin films deposited at $T_s = 250$ °C on the quartz glass substrate are presented in Fig. 1a reveals the optical absorption properties of V_2O_5 thin films measured from 300 to 900 nm. The absorption spectra indicate the presence of a sharp peak at 316 nm specifying the strong UV absorption of V_2O_5 thin films. On the other hand, Fig. 1b discloses the transmittance spectra of the thin films obtained in the range from 300 to 900 nm. The transmittance observed from the figure shows the variation from 20% at 300 nm to 80% at 800 nm in the visible spectrum range. The optical bandgap of V_2O_5 films was evaluated using a well-known Tauc's plot as depicted in Fig. 1c. The optical property of V_2O_5 thin films is well-defined using direct bandgap transition by drawing a Tauc's plot between $(\alpha h\nu)^2$ versus photon energy (E). The extrapolation of the linear portion of the data at the transition edge towards to zero absorption evaluates the optical bandgap. The obtained optical bandgap of V_2O_5 films is found to be 3.28 eV, which was found similar with the earlier reports [28, 30, 34]. Observed sensible transmittance along with a larger

bandgap makes V_2O_5 thin films suitable for various optical and optoelectronic applications [23].

Structural properties of the V_2O_5 films deposited at $T_s = 250$ °C on glass and Ge substrates were evaluated using Raman and XRD measurements as shown in Fig. 2a depicts the XRD measurements of V_2O_5 /Ge layers illustrated with various crystal orientations of V_2O_5 thin film. The XRD peaks are found to be very sharp indicating the higher crystallinity nature of thin films. The predominant peaks observed in the XRD spectrum are well matched with the JCPDS data (vide code No.: 00-041-1426, 01-075-0457), which indicates the deposited V_2O_5 thin films are in orthorhombic structure [26]. The maximum diffraction observed in the spectrum are marked with different orientations of V_2O_5 crystals like (101), (011), (120), (421), (712), (821) and (132) along with the Ge substrate peak. This evidences the formation of V_2O_5 thin film on n-Ge with orthorhombic crystal structure. The strain distribution model as proposed by Abdel-Rahman and Sabry [35] is applied to the peaks shown in the XRD spectrum to evaluate average values of crystallite size, microstrain, dislocation density and was found to be 93.41 nm, 0.77×10^{-3} , 1.04×10^{15} lin. m⁻², respectively. In addition, Raman measurements of V_2O_5 films deposited on quartz glass were also performed to confirm the structural properties of the films. The Raman vibrational spectrum (shown in Fig. 2b) evaluated from 100 to 1000 cm⁻¹ indicates different characteristic bands. The vibrational bands observed in the Raman spectrum also confirm the orthorhombic crystal nature of V_2O_5 films [36]. A highly intense distinct vibrational band is observed at 144 cm⁻¹ attributed to V–O–V chains usually corresponding to the layered growth of V_2O_5 films with orthorhombic structure. The peak at 993 cm⁻¹ might be associated with V=O bond stretching mode which indicates the crystalline quality and stoichiometry of films [37]. The other vibrational bands such as 526 cm⁻¹ and 700 cm⁻¹ correspond to the V–O phonon bond and V–O–V bonds, respectively, [38]. These measurements confirm the formation of highly crystalline orthorhombic V_2O_5 thin films on n-Ge.

The surface morphology of the V_2O_5 thin films deposited at a substrate temperature of 250 °C on n-Ge is analysed using two different approaches AFM and FESEM. The morphology of V_2O_5 thin films on n-Ge using AFM measurements is shown in Fig. 3a and b represent the 2D and 3D micrographs of V_2O_5 thin films with the active scanning area of 5 µm × 5 µm. The surface smoothness of the thin films is analysed

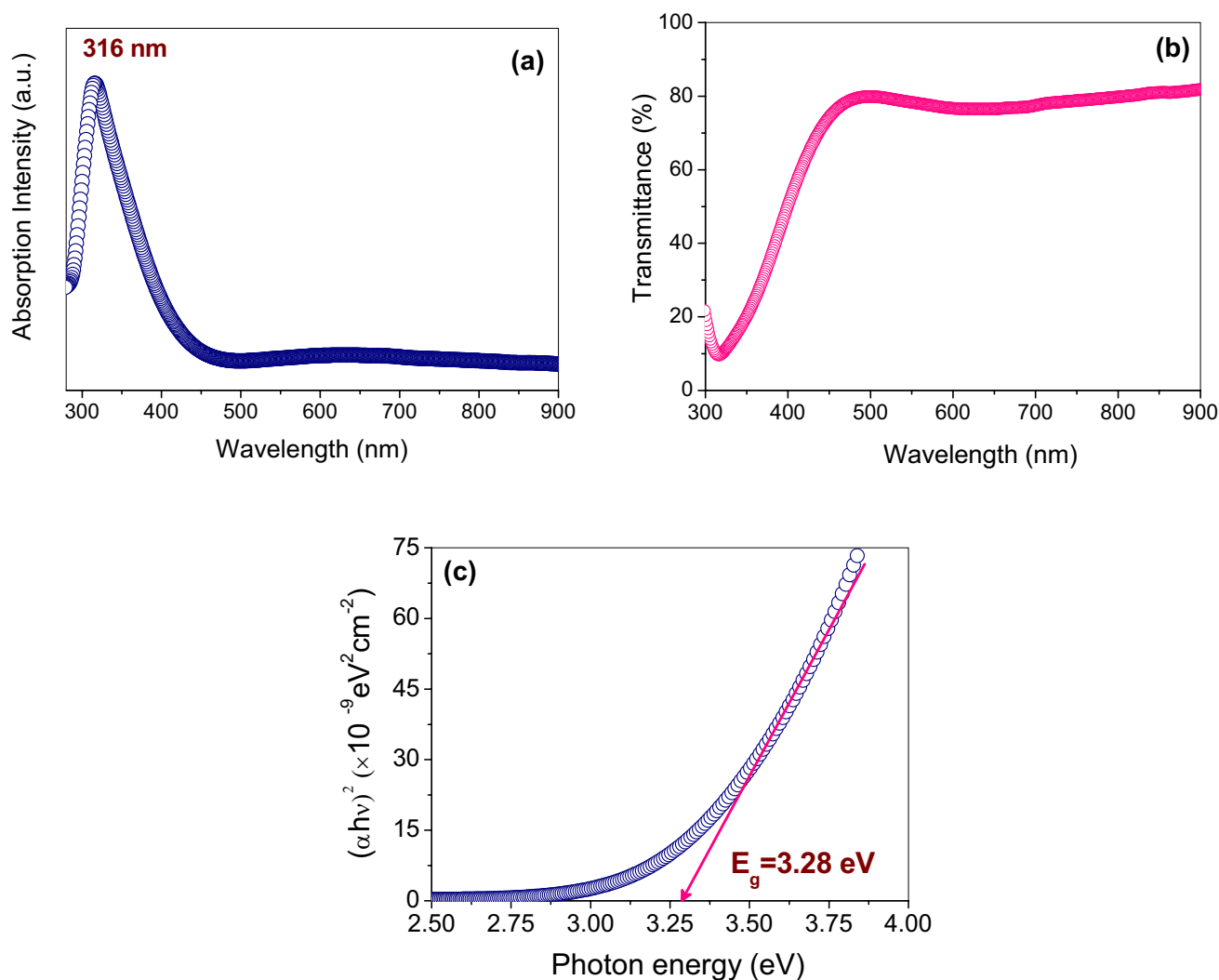


Fig. 1 Optical Properties of V_2O_5 thin films deposited at substrate temperature of 250 °C. **a** Absorption spectrum. **b** Transmittance and **c** Tauc's plot between $(\alpha h\nu)^2$ versus photon energy

using RMS roughness which was found to be 2.44 nm. The magnitude of roughness indicates the surface is found to be smooth with a uniform distribution of crystallites. On the other hand, the surface analysis of V_2O_5 thin films and interface analysis of $\text{V}_2\text{O}_5/\text{Ge}$ layers is also carried out using FESEM and cross-sectional FESEM measurements, respectively. Surface analysis of V_2O_5 thin films along with EDS measurements (not presented here) is revealed in Fig. 4a. The surface of the film shows diffused grains with smooth morphology and confirms adequate at% of constituent elements of the V_2O_5 thin films which are shown in the inset of Fig. 4a, b illustrates the interface analysis of $\text{V}_2\text{O}_5/\text{Ge}$ layers using cross-sectional FESEM measurements with a bar at 400 nm. Well-defined structured

layers of V_2O_5 and Ge were identified with ~ 68 nm thickness of V_2O_5 layer. Further, a uniform thickness of the V_2O_5 layer is observed throughout the measured micrograph. This analysis indicates the formation of a highly stable interface between V_2O_5 and Ge layers.

3.2 Electrical and photo-electrical properties

The efficacy of Schottky devices was primarily analysed using electronic properties which might be obtained using current–voltage (I – V) measurements. Electronic properties of Au/Ge (reference) and $\text{V}_2\text{O}_5/\text{Ge}$ heterojunction are presented in Fig. 5 at room temperature under dark conditions. Figure 5a reveals the I – V properties of reference contact and

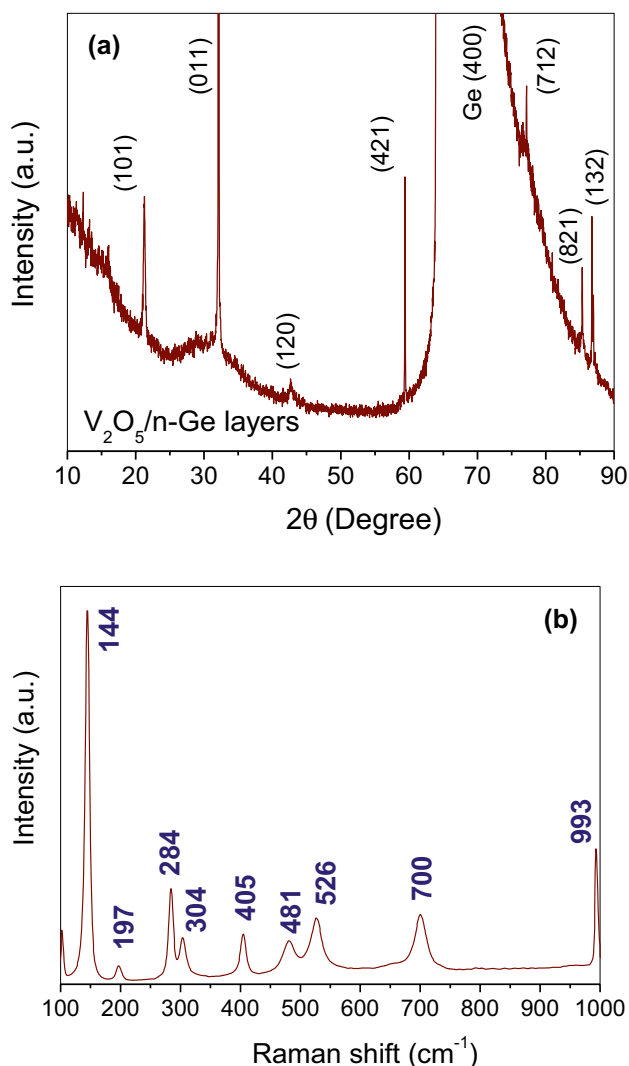


Fig. 2 Structural properties of V_2O_5 thin films. **a** XRD spectrum of V_2O_5 /Ge layers. **b** Raman measurements of V_2O_5 thin films on glass substrate

heterojunction. The barrier formation across the metal-Ge and V_2O_5 -Ge interface can be characterized using standard thermionic emission (TE) theory. The current across any junction based on TE theory is represented as [39],

$$I = I_0 \left[\exp \left(\frac{q(V - IR_s)}{nkT} \right) - 1 \right] \quad (1)$$

where saturation current is defined as,

$$I_0 = AA^*T^2 \exp \left(\frac{-q\Phi_b}{kT} \right) \quad (2)$$

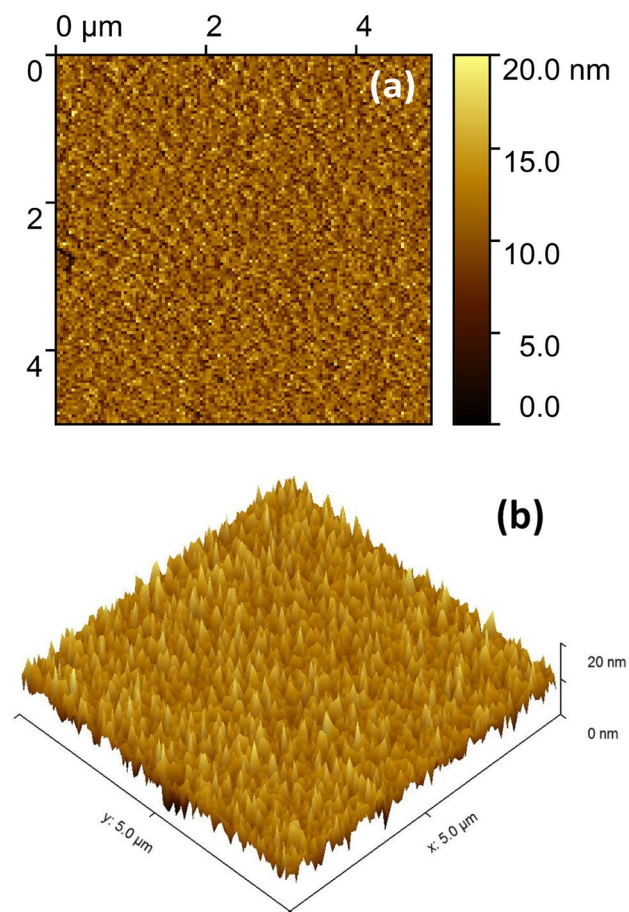


Fig. 3 Surface topography of V_2O_5 /Ge layers using AFM measurements. **a** 2D-image with scanning area of $5 \mu m \times 5 \mu m$. **b** 3D micrographic image

The terms involved in the above equations have the usual meaning defined elsewhere [40]. The barrier parameters such as barrier height (Φ_b) and ideality factor (n) were evaluated based on the extraction of saturation current I_0 from logarithmic current vs. voltage plot. The reverse saturation current as noted in Fig. 5a indicate decreased magnitude for the V_2O_5 /Ge heterojunction than without the V_2O_5 layer (IL). The magnitudes of Φ_b and n of the Au/Ge contact were found to be 0.54 eV and 2.02, whereas for the heterojunction is 0.61 eV and 2.16, respectively. The larger ideality factor observed in both the contacts indicates the valid current transport process other than thermionic emissions such as tunneling or diffusion. The other reasons for the deviation of n from unity include the presence of interface states, the influence of series resistance and carrier recombination and generation through deep states at the junction [41, 42]. The value of Φ_b of the

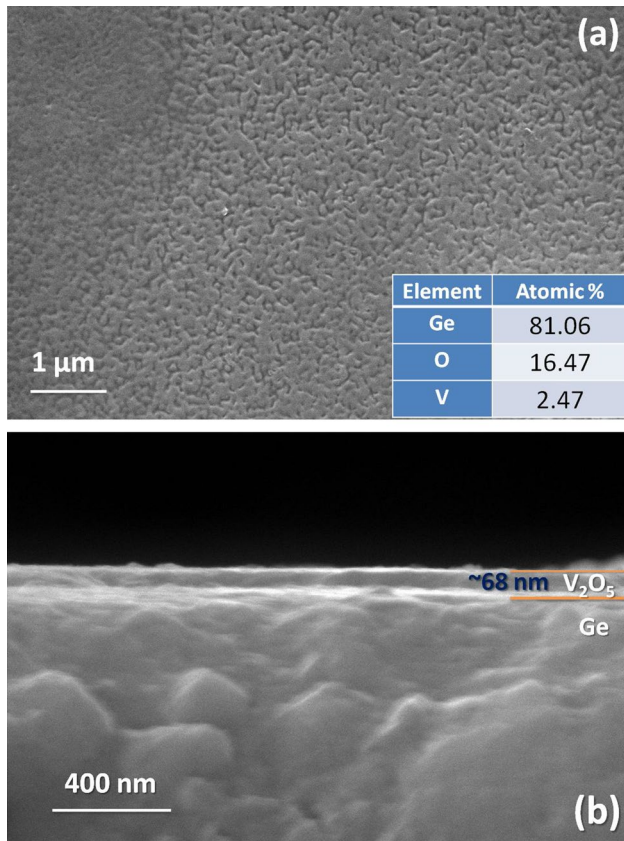


Fig. 4 FESEM analysis of V_2O_5/Ge layers. **a** Morphology of V_2O_5 surface with EDS elemental analysis provided at the inset. **b** Cross-sectional FESEM image of V_2O_5/Ge layers

heterojunction was found to be increased significantly than without IL. The barrier parameters such as barrier height and ideality factor for $V_2O_5/n\text{-Ge}$ heterojunction were found comparable with earlier reports of similar structure as depicted in Table 1.

The enhancement of barrier height in the $V_2O_5/n\text{-Ge}$ heterojunction than without V_2O_5 interfacial layer signifies the tuning of band alignments, passivation of states and creation of an inversion layer with dipoles at the $V_2O_5\text{-Ge}$ interface [15, 17]. The non-ideal current–voltage characteristic behavior can be attributed to the greater interface state density. Passivating the dangling bonds is an important result of putting an oxide layer on the semiconductor surface. This passivation can reduce the anomalies in the diode current–voltage characteristic behavior by minimizing the surface states. Achieving a low leakage current is another crucial component of the oxide interfacial layer between the MS Schottky diode [44]. The V_2O_5 oxide layer also serves a dual purpose of reducing

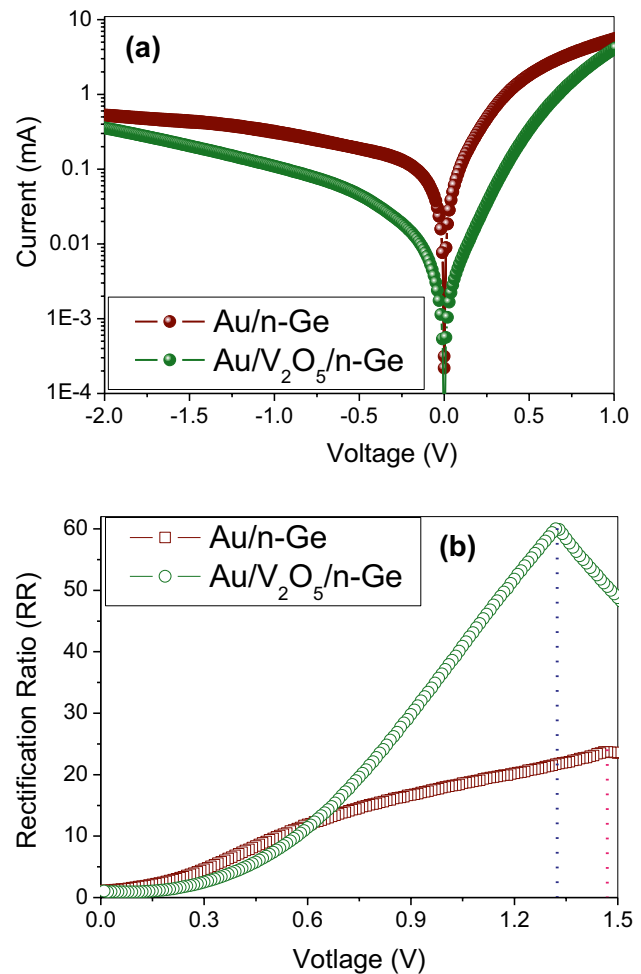
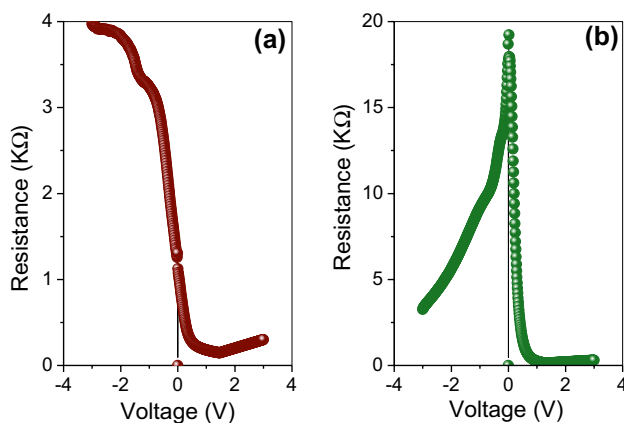


Fig. 5 Electrical measurements of V_2O_5/Ge heterojunction. **a** I - V properties of Au/Ge contact and heterojunction. **b** Rectification ratio of the contacts versus bias potential

the interface charge and improving the band-offset between the overlying passivation layer and the Ge semiconductor, which in turn lowers the tunneling leakage currents. Further, the V_2O_5 layer acts as a barrier which prevents the inter-diffusion between metal to Ge substrate which predominantly improves the device performance. A similar observation in devices with the V_2O_5 interlayer was also made by Balasubramani et al., Kaya et al. and Mahato et al. [28, 41, 42]. Both the contacts show better rectifying characteristics as evidenced from Fig. 5a. The rectification properties of the contacts can be analysed using a parameter defined as rectification ratio (RR) which is the ratio of forward current over reverse current for the fixed bias potential ($RR = (I_F/I_R)_{V=\text{const}}$). The variation of RR for the bias potential is shown in Fig. 5b for the contacts

Table 1 Review of published studies on the improved barrier height in metal/n-Ge Schottky diodes that utilize an interfacial layer (IL)

Structure	Schottky barrier parameters				References
	Φ_b (eV)	n	R_S (Ω)		
			($dV/d\ln I$) Vs I	$H(I)$ Vs. I	
Au/Graphene/n-Ge	0.63	1.30	3.15	3.16	Khurelbaatar et al. [43]
Ni/TaO(5 nm)/n-Ge	0.56	1.15	14.9	15.3	Lee et al. [32]
Pt/ZnO/n-Ge	0.65	2.17	46	46	Kim et al. [33]
Au/WO ₃ /n-Ge	0.63	1.1	17	17	Thomas et al. [16]
Au/V ₂ O ₅ /n-Ge	0.61	2.16	199	193	Present work

**Fig. 6** Junction resistance (R_j) versus bias potential plot to determine R_s and R_{sh} for **a** Au/Ge contact and **b** V₂O₅/Ge heterojunction

with and without IL. The RR for both the contacts seems to be increasing with the bias indicating the junction is composed of the inhomogeneous barrier [45]. The RR of V₂O₅/Ge heterojunction was found to be larger by four times than the contacts without IL. A maximum RR for the heterojunction is obtained at $V = 1.32$ and $V = 1.46$ for the contacts with and without IL respectively.

In the design of solar/photovoltaic cells, two parameters which are very much influencing the efficiency of the cells are series resistance (R_s) and shunt resistance (R_{sh}). The effectiveness of the cells is significantly larger for the low magnitude of R_s and high magnitude of R_{sh} . These parameters are evaluated in the forward bias I - V properties of the contacts by taking a plot between junction resistance (R_j) versus bias voltage as shown in Fig. 6a and b. The magnitudes of R_s and R_{sh} for the Au/Ge contact were found to be 147 Ω and 1.125 k Ω whereas for the heterostructure is 133 Ω and 17.825 k Ω , respectively. The extracted values of R_s and R_{sh} indicate a

slight decrease in R_s and a notable increase in R_{sh} for the heterostructure compared to the Au/Ge contact. In general larger magnitudes of R_{sh} and smaller R_s magnitudes appreciably enhance the performance of solar cell devices [46]. This shows the usefulness of utilizing the V₂O₅ layers in the design of photovoltaic/solar cells.

Considering the forward bias current-voltage properties of the contacts, there exists a linear variation of current and voltage for the smaller magnitudes of voltage (till for $V < 2$ kT/q). As the bias potential increases ($V > 3$ kT/q), the influence of series resistance (R_s) starts to decrease the magnitude of the current. The evaluation of R_s can make use of the range of bias potential that results in the deviation of linearity in diode current. A standardized approach in the evaluation of R_s in the non-linear portion of forward-biased I - V properties of the contact was proposed by Cheung and Cheung [47]. In this approach, Cheung's functions are utilized to validate the barrier parameters such as R_s , Φ_b and n with other approaches. The Cheung's functions are defined as,

$$\text{Cheung function1 : } \frac{dV}{d(\ln I)} = IR_s + n \left(\frac{kT}{q} \right) \quad (3)$$

$$\begin{aligned} \text{Cheung function2 : } H(I) &= V - n \left(\frac{kT}{q} \right) \ln \left(\frac{I}{AA^*T^2} \right) \\ &= n\Phi_b + IR_s \end{aligned} \quad (4)$$

The terms defined here in the functions have usual meanings [40]. A variation of $dV/d(\ln I)$ versus forward current of both the contacts is shown in Fig. 7a and b to determine the barrier parameters such as R_s and n . The magnitudes of R_s and n as determined from Cheung's function1 for the Au/Ge

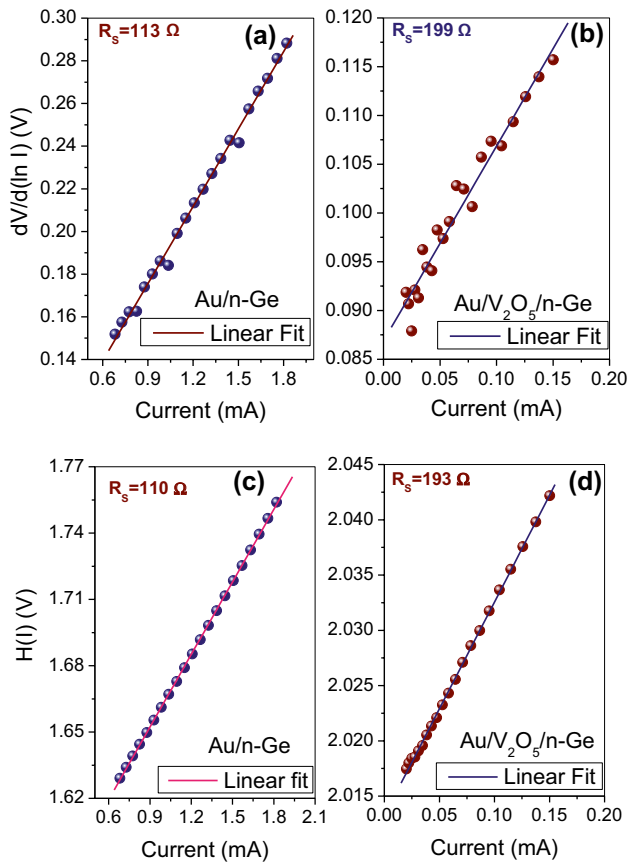


Fig. 7 Cheung's plot of contacts (a and b) $dV/(d \ln I)$ versus forward current of Au/Ge contacts and heterojunction, (c and d) Cheung's function $H(I)$ versus forward current of Au/Ge contacts and heterojunction

contact were found to be 113Ω and 3.01 , whereas for the heterojunction is observed as 199Ω and 3.41 . The magnitude of n obtained from Cheung's function1 is utilized in the evaluation of $H(I)$ (Cheung's function2). The extracted value of $H(I)$ is plotted against the forward current for both the contacts is shown in Fig. 7c and d to evaluate the magnitudes of R_s and Φ_b from the slope and intercept of the linear fit of data. The obtained values of R_s and Φ_b using Cheung's function2 for the Au/Ge contact are 110Ω and 0.52 eV and for heterojunctions are 193Ω and 0.59 eV, respectively. The values of R_s as determined from Cheung's functions show similarity with each other and are also found identical with the R_s obtained from the junction resistance plot. On the other hand, Φ_b values evaluated from Cheung's function2 matches well with the same from $\ln(I)$ versus V plot based on TE theory.

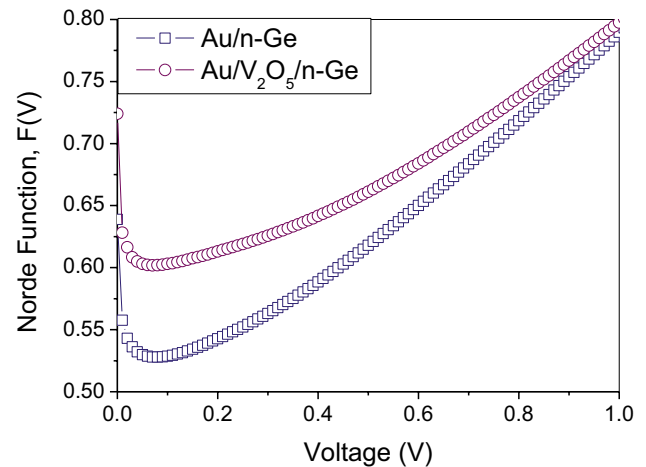


Fig. 8 Norde function $F(V)$ versus bias potential of Au/Ge contact and the heterojunction

Norde also proposed another alternate approach to evaluate the barrier parameters such as Φ_b and R_s using Norde function defined as [48],

$$F(V) = \frac{V}{\gamma} - \frac{q}{kT} \ln \left(\frac{I(V)}{AA^*T^2} \right) \quad (5)$$

The terms involved in the above equation have the usual meaning [40]. A plot of Norde function $F(V)$ determined from Eq. (5) versus voltage is shown in Fig. 8, which is used to determine the barrier parameters such as Φ_b and R_s . The minimum magnitude of the Norde function [$F(V_{\min})$] from the plot can tell about the minimum value of voltage ($V_{\min}$) and current ($I_{\min}$). By using these quantities the barrier parameters are defined as [40],

$$\Phi_b = F(V_{\min}) + \frac{V_{\min}}{\gamma} - \frac{kT}{q} \quad \text{and} \quad R_s = \frac{kT(\gamma - n)}{qI_{\min}} \quad (6)$$

The extracted values of Φ_b and R_s from the Norde plot for the Au/Ge contact and V_2O_5/Ge heterojunction were found to be 143Ω , 0.53 eV and 2174Ω , 0.6 eV for the heterojunction, respectively. The magnitude of barrier height of both the contacts evaluated using $F(V)$ shows its similarity with other approaches indicating larger consistency [41]. The series resistance of the Au/Ge contacts was found alike with the earlier approaches. Whereas the series resistance of the heterojunction obtained using the Norde plot was found significantly higher than other approaches due to the region dependence in the evaluation of series resistance and non-ideal nature of the contact [49]. The

comparison of the barrier parameters of the Au/Ge and heterojunction contacts indicates the approaches adopted are consistent and show their validity.

Current transport mechanisms involved in the forward bias I - V measurements of the contacts are truly defined by the space charges at the junction due to either bulk/thermally injected carriers. Migration of carriers across the junction is also a bias-dependent process. Variation of forward bias potential causes to exhibit different transport processes such as space charge limited current (SCLC) conduction with or without the influence of traps. The current produced due to the transport of carriers across the junction is related to the bias that satisfies the power law defined as $I \propto V^m$. The magnitude of m gives the details of dominant current transport processes with bias. The magnitude of $m \sim 1$ indicates the current maintains a linear relation with voltage termed as ohmic conduction. On the other hand, the range of m between 1 and 2 indicates the thermally generated carriers transported across the junction produce space charges termed as the SCLC conduction process. For all the magnitudes of m larger than 2 shows that the bulk injection of carriers dominates the thermally generated carriers and the possible conduction mechanism is associated with SCLC assisted with the traps. In the present investigation, the evaluation of the magnitude of m is done by plotting a graph between $\log$ (current) vs. $\log$ (voltage). The slope of the linear portion of the plot reveals the magnitude of m . A double logarithmic plot of current versus voltage is shown in Fig. 9 for the Au/

Ge and heterojunctions. The figure reveals two linear regions for the Au/Ge contacts termed region-A and region-B and the value of m is noted in the graph for each linear portion. The value of m found in region-A signifies a linear variation of current with voltage indicating ohmic conduction is dominant for the voltage range from 0.01 to 0.15 V. As the bias voltage increases the magnitude of m seems to be increasing revealing the dominance of bulk carrier injection and the possibility of SCLC transport process. On the other hand, heterostructure shows three linear portions marked as region-I, region-II and region-III with an increased magnitude of m . At low bias potential ($0.01 < V < 0.1$) the value of m affirms a linear relation between current and voltage indicating ohmic conduction is the dominant conduction mechanism. As the bias potential is increased from 0.1 V to 0.26, the value of m increased to 1.85 which shows the activation of the SCLC transport mechanism. Finally, with the bias voltages ($V > 0.26$) the slope increases to a value equal to 3.54 illustrating that carrier conduction is mainly dominated by SCLC associated with the traps. Similar observations like trap-assisted SCLC process in V_2O_5 /Si heterojunction at higher bias potential were reported by Ali [50].

Frequency-dependent series capacitance and impedance properties of the Au/Ge contacts and V_2O_5 /Ge heterojunction were studied at variable bias voltage range from 0.25 V to 1 V in steps of 0.25 V (shown in Fig. 10). Figure 10a describes the frequency-dependent capacitance for fixed bias voltages for V_2O_5 /Ge heterojunction. The figure reveals a larger dispersion of capacitance at low frequencies than at high frequencies for the increased bias voltages. Further, an observed decreased capacitance with increased frequency indicates the influence of surface states on depletion capacitance. At high frequencies, the interface states experience a very short carrier lifetime than at low frequencies. This generally makes the carriers unable to follow the AC frequency signal at higher frequencies than at low frequencies which causes them to exhibit larger capacitance at lower frequencies than at high frequencies [51, 52]. Similarly, Fig. 10c shows the variation of capacitance with frequency for different bias voltages for the Au/Ge contact. The figure indicates a very small dispersion of capacitance at all frequencies for different bias voltages. This needs a detailed investigation to look into the possible reasons for the smaller dispersion of capacitance for the Au/Ge contact. Variation of series resistance with frequency

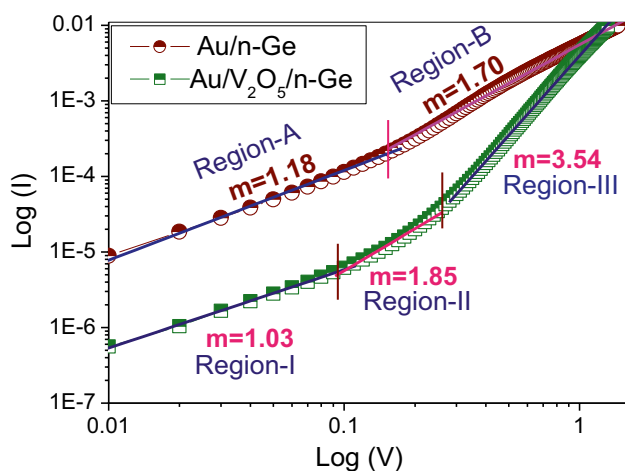


Fig. 9 Forward current conduction processes of contacts using a plot between double logarithmic current versus voltage

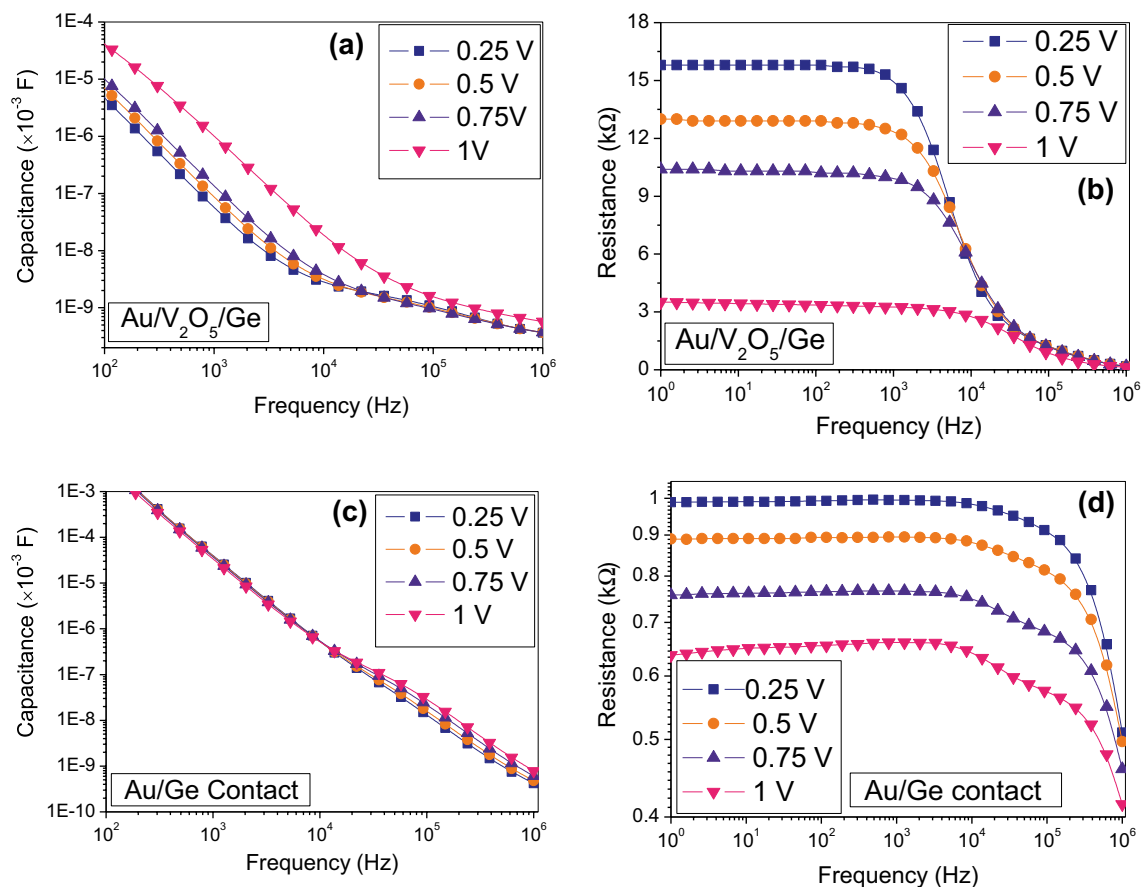


Fig. 10 Frequency-dependent capacitance and series resistance at varied bias potential for the $\text{V}_2\text{O}_5/\text{Ge}$ heterojunction (a and b) and Au/Ge contact (c and d)

at different bias voltages for the $\text{V}_2\text{O}_5/\text{Ge}$ heterojunction and Au/Ge contact is presented in Fig. 10b and d, respectively. The series resistance of the contacts found to be decreased with increasing frequency and similar response is for all bias potentials. Larger series resistance is evidenced at smaller frequencies and decreased resistance at larger frequencies which indicates the influence of interface states with frequency. The magnitude of series resistances in heterojunction is found to be increased at all frequencies and bias potentials than the Au/Ge contacts which might be associated with the presence of V_2O_5 interlayer (IL) and the particular distribution of interface states at the IL-Ge junction [52].

Photo-electrical measurements were conducted for the $\text{V}_2\text{O}_5/\text{Ge}$ heterojunction under illumination at 100 mW cm^{-2} using a solar simulator operated at 1 sun condition. The electrical measurement of heterojunction under illumination is presented in Fig. 11a. The results reveal that the photocurrent is exhibited by

the heterojunction under reverse biased condition. Crucial parameters to analyze the efficiency of the $\text{V}_2\text{O}_5/\text{Ge}$ heterojunction under illumination might be photo-to-dark current ratio (PDCR) also termed as sensitivity and photoresponsivity. The detailed evaluation procedure of these parameters of photovoltaic cells can be found elsewhere [53]. Figure 11b defines the sensitivity and photoresponsivity properties of the heterojunction under illumination. The sensitivity parameter of the heterojunction was found to be increasing with reverse bias potential and found maximum at -0.4 V (PDCR = 2.1%) and decreases thereafter. On the other hand, photoresponsivity was found to be increasing with bias potential and at -1 V the responsivity seems to be 534 mA W^{-1} . The photoresponsivity observed for the heterojunction seems to be comparable with earlier reports [26, 28, 54, 55]. Applying a reverse bias increases the width of the depletion junction producing an increased

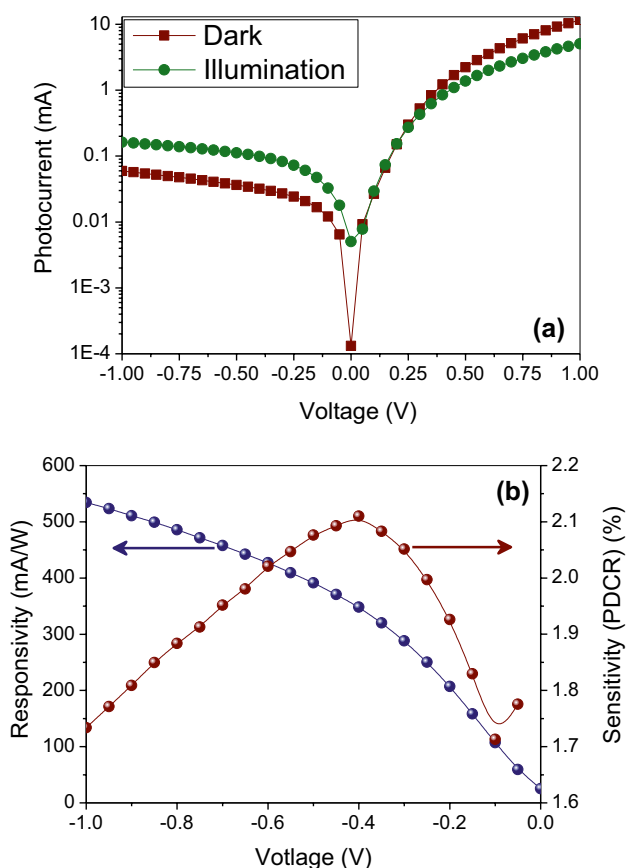


Fig. 11 Photovoltaic properties of V_2O_5/Ge heterojunction. **a** I – V properties under dark and illumination at 100 mW cm^{-2} in 1 sun condition. **b** Variation of PDCR and photoresponsivity of the heterojunction under illumination versus reverse bias potential

responsivity. Operating under these conditions does tend to produce a larger dark current, but this can be limited based upon the photodiode material. Further, the increase in the responsivity of V_2O_5/Ge heterojunction in general was because of the presence of the oxide film between the semiconductor and metal which could improve the contact properties and reduced the interface conditions. This in turn reduces the process of recombination in a way that would increase the photocurrent and hence increases the responsivity of the heterojunction [56]. The observation of the photovoltaic behavior of the V_2O_5/Ge heterojunction initiates the development of Ge-based heterojunction solar cells to new heights using oxide-based interlayers. Though the efficiencies of these heterojunctions are not remarkable, there is a scope to improve the performance

of the cells by doping or altering the post and pre-deposition parameters like annealing.

4 Conclusion

A detailed investigation of the physical properties of the V_2O_5 thin films deposited at 250°C is performed using optical, structural and morphological measurements. Optical properties of the films were presented with absorption and transmittance properties where maximum absorption is absorbed in 316 nm and transmittance was increased to 80% in the visible region wavelength. The direct transition optical bandgap of V_2O_5 films was found to be 3.28 eV. Structural properties of films were studied using XRD and Raman measurements and the result indicates a larger magnitude of crystallinity with orthorhombic structure and high crystalline quality with characteristic vibrational bands of V_2O_5 . The V_2O_5 film surface was found to be very smooth with an RMS roughness of 2.44 nm as evidenced by AFM measurements. The electrical measurements of Ge contacts with and without V_2O_5 interlayer disclose superior barrier properties such as larger barrier height, larger shunt resistance and noticeable rectification ratio for the V_2O_5/Ge heterojunction than without interlayer. The barrier parameters evaluated using different approaches show their consistency and validity. The frequency-dependent series capacitance and resistance values were evaluated for fixed bias potential. The series capacitance shows larger dispersion at low frequency than at high frequency indicating the influence of interface states at the V_2O_5 –Ge. The photovoltaic properties of V_2O_5/Ge heterojunction under illumination at 100 mW cm^{-2} show increased photoresponse with increased reverse bias voltage. The responsivity of the heterojunction is found to be 534 mA W^{-1} at -1 V and maximum sensitivity is observed as 2.1% at -0.4 V . These results initiate a pathway to develop highly efficient Ge-based heterojunction photovoltaic cells using transition metal oxides such as V_2O_5 as an interlayer.

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Author contributions

Conceptualization, Henry Thomas G and Ashok Kumar A; Data curation, Ashok Kumar A, Rajagopal Reddy V and Kaleemulla S; Formal analysis, Kaleemulla S and Ashok Kumar A; Methodology, Henry Thomas G, Ashok Kumar A, and Rajagopal Reddy V; Resources, Kaleemulla S, and Rajagopal Reddy V; Supervision, Ashok Kumar A; Validation, Ashok Kumar A, Kaleemulla S and Rajagopal Reddy V; Visualization, Henry Thomas G and Ashok Kumar A; Writing – original draft, Henry Thomas G and Ashok Kumar A; Writing – review & editing, Ashok Kumar A, Kaleemulla S and Rajagopal Reddy V. All authors have read and agreed to the published version of the manuscript.

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Data availability

Where no new data were created, or where data are unavailable due to privacy or ethical restrictions. The authors are herewith giving the consent that the data be made available on request.

Declarations

Conflicts of interest The authors confirm that there is no conflict of interest or competing interest about this work with anybody.

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